

In the event of end-stage heart failure, a left ventricular assist device (LVAD) can be used as a heart transplant bridge to keep a patient alive. The first iterations of LVADs were pulsatile and mimicked the physiological pumping action of the heart (Figure 1).

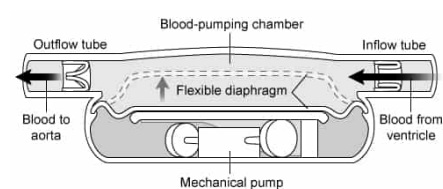


Figure 1 A pulsatile-flow LVAD schematic

A pulsatile-flow LVAD assists ventricular systole by mechanically pumping blood from a weakened left ventricle into the aorta through a pair of one-way valves. The pressure differential ΔP generated by the pump is related to cardiac output (CO) and vascular resistance (VR):

$$\Delta P = CO \times VR$$

Equation 1

The efficiency and performance of the heart (or LVAD) can be determined by the patient's cardiac pressure-volume (PV) loop. A cardiac PV loop plots the pressure and volume of the blood in the left ventricle throughout a single cardiac cycle (Figure 2).

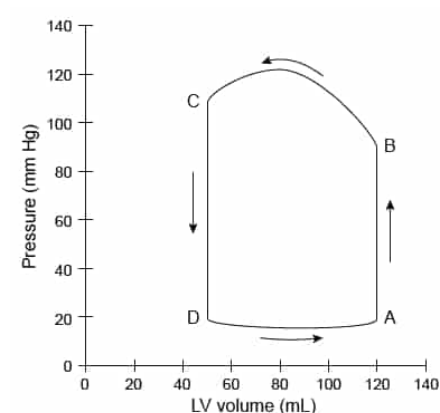


Figure 2 Cardiac PV Loop of a pulsatile-flow LVAD

Blood pressure is often represented by only two numbers: arterial systolic and diastolic pressures. A more detailed representation of blood pressures is shown in a blood pressure profile, which graphs the blood pressure throughout the length of the different vessel groups (Figure 3). Multiple pressure fluctuations are shown within a vessel because the pressure is traced across multiple cardiac cycles.

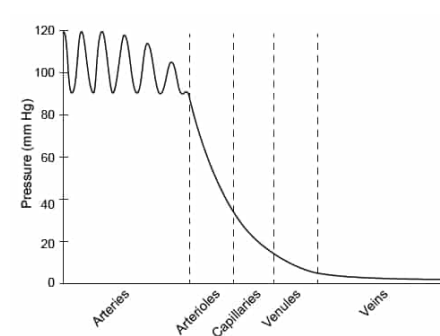


Figure 3 Blood pressure profile of the different vessels

The mean arterial pressure (MAP) is used to approximate the average blood pressure of an individual's arteries:

$$MAP = DP + (SP - DP)/3$$

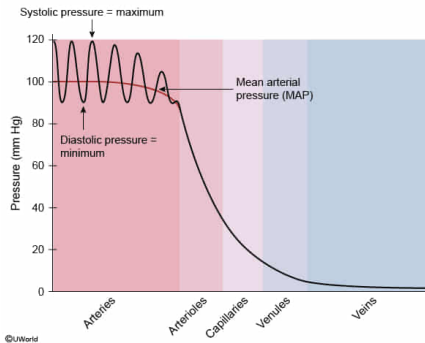
where DP is diastolic pressure and SP is systolic pressure. What is the MAP for the blood pressure profile shown in Figure 3?

- ☐ A. 40 mmHg [13%]
☐ B. 80 mmHg [8%]
☒ C. 100 mmHg [69%]
☐ D. 105 mmHg [8%]

Correct

69% answered correctly

Explanation:



Mean arterial pressure (MAP) is defined as the average arterial blood pressure throughout a cardiac cycle. Due to the oscillatory nature of arterial blood pressure (as seen in the figure above), MAP cannot simply be calculated as the midpoint between systolic and diastolic pressure (**Choice D**). MAP is approximated from arterial systolic pressure (SP) and diastolic pressure (DP) with the given equation

$$MAP = DP + \frac{(SP - DP)}{3}$$

Arterial **systolic** pressure is the **maximum** blood pressure in the arteries due to ventricular contraction, and arterial **diastolic** pressure is the **minimum** blood pressure in the arteries due to heart relaxation. From Figure 3, the highest arterial pressure is 120 mmHg, and the lowest arterial pressure is 90 mmHg.

Using 120 mmHg as SP and 90 mmHg as DP ,

$$MAP = 90 \text{ mmHg} + \frac{(120 \text{ mmHg} - 90 \text{ mmHg})}{3} = 100 \text{ mmHg}$$

The MAP of the blood pressure profile is **100 mmHg**.

(Choice A) The systolic and diastolic measurements for MAP are taken from the arteries. If the lowest blood pressure in the entire circulatory system (0 mmHg at the veins) is used as the arterial diastolic pressure, MAP will be incorrectly calculated as 40 mmHg.

Educational objective:

Systolic pressure is the maximum measured blood pressure due to ventricular contraction, and diastolic pressure is the minimum measured blood pressure due to heart relaxation. The mean arterial pressure (MAP) is approximated from arterial systolic (SP) and diastolic (DP) pressures by $MAP = DP + \frac{(SP - DP)}{3}$.

Time spent: 0 sec

Subject:
Physics

QID: 400819

System:
Fluids

In the event of end-stage heart failure, a left ventricular assist device (LVAD) can be used as a heart transplant bridge to keep a patient alive. The first iterations of LVADs were pulsatile and mimicked the physiological pumping action of the heart (Figure 1).

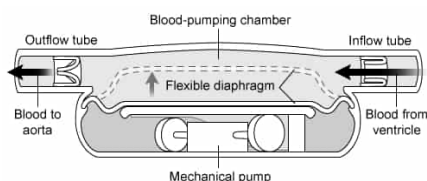


Figure 1 A pulsatile-flow LVAD schematic

A pulsatile-flow LVAD assists ventricular systole by mechanically pumping blood from a weakened left ventricle into the aorta through a pair of one-way valves. The pressure differential ΔP generated by the pump is related to cardiac output (CO) and vascular resistance (VR):

$$\Delta P = CO \times VR$$

Equation 1

The efficiency and performance of the heart (or LVAD) can be determined by the patient's cardiac pressure-volume (PV) loop. A cardiac PV loop plots the pressure and volume of the blood in the left ventricle throughout a single cardiac cycle (Figure 2).

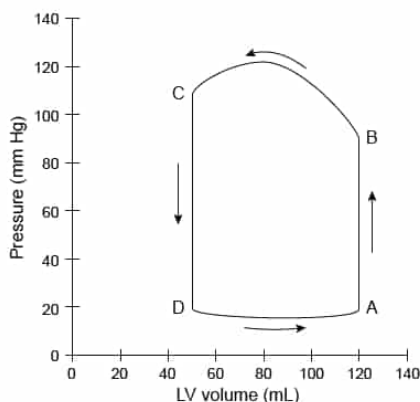


Figure 2 Cardiac PV Loop of a pulsatile-flow LVAD

Blood pressure is often represented by only two numbers: arterial systolic and diastolic pressures. A more detailed representation of blood pressures is shown in a blood pressure profile, which graphs the blood pressure throughout the length of the different vessel groups (Figure 3). Multiple pressure fluctuations are shown within a vessel because the pressure is traced across multiple cardiac cycles.

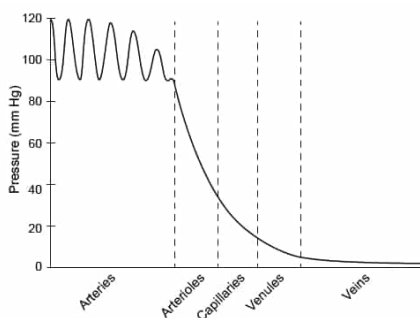


Figure 3 Blood pressure profile of the different vessels

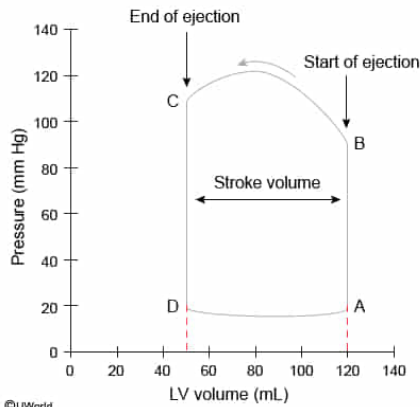
If the pulsatile-flow LVAD in Figure 2 pumps with a frequency of 1.5 Hz, what is the volumetric flow rate of blood throughout the systemic circulatory system?

- ☐ A. 2,800 mL/min [12%]
- ✓ ☒ B. 6,300 mL/min [59%]
- ☐ C. 9,900 mL/min [19%]
- ☐ D. 10,800 mL/min [8%]

Correct

59% answered correctly

Explanation:



The **volumetric flow rate** of blood throughout the systemic circulatory system is determined by the heart (or LVAD) and is **equal to the cardiac output**. Cardiac output is the volume of blood pumped by the heart every minute and is the product of stroke volume and heart rate:

$$\text{cardiac output} = (\text{stroke volume}) \times (\text{heart rate})$$

Stroke volume is the volume of blood that is ejected from the left ventricle (LV) in one cardiac cycle. On a cardiac PV Loop (Figure 2), ejection occurs when LV volume is decreasing (BC curve). Therefore, stroke volume is the **difference in LV volume** between the **beginning and end of ejection**. From the PV loop, the LV volume at the start and end of ejection is 120 mL and 50 mL, respectively. Therefore, the stroke volume is:

$$120 \text{ mL} - 50 \text{ mL} = 70 \text{ mL}$$

Heart rate is the number of cardiac cycles per minute. The given frequency of 1.5 Hz is converted to cycles/min using dimensional analysis:

$$\frac{1.5 \text{ cycles}}{1 \text{ s}} \times \frac{60 \text{ s}}{1 \text{ min}} = 90 \frac{\text{cycles}}{\text{min}}$$

Using a stroke volume of 70 mL and a heart rate of 90 beats/min:

$$70 \text{ mL} \times 90/\text{min} = 6,300 \text{ mL/min}$$

The cardiac output and volumetric flow rate of blood is **6,300 mL/min**.

(Choice A) The SI unit of frequency, hertz (Hz), is equal to $\frac{1 \text{ cycle}}{1 \text{ s}}$. If 1.5 Hz is incorrectly used as $\frac{1 \text{ cycle}}{1.5 \text{ s}}$, cardiac output would be incorrectly calculated as 2,800 mL/min.

(Choice C) If the pressure at the end of ejection (110 mmHg) is incorrectly used as stroke volume, cardiac output would be incorrectly calculated as 9,900 mL/min.

(Choice D) If the maximum LV volume (120 mL) is incorrectly used as stroke volume, cardiac output would be incorrectly calculated as 10,800 mL/min.

Educational objective:

The volumetric flow rate of blood throughout the systemic circulatory system is equal to the cardiac output of the heart (stroke volume $\times$ heart rate). On a cardiac PV loop, stroke volume is the difference in volume between the beginning and end of ejection.

In the event of end-stage heart failure, a left ventricular assist device (LVAD) can be used as a heart transplant bridge to keep a patient alive. The first iterations of LVADs were pulsatile and mimicked the physiological pumping action of the heart (Figure 1).

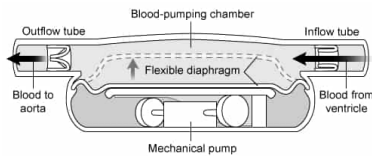


Figure 1 A pulsatile-flow LVAD schematic

A pulsatile-flow LVAD assists ventricular systole by mechanically pumping blood from a weakened left ventricle into the aorta through a pair of one-way valves. The pressure differential ΔP generated by the pump is related to cardiac output (CO) and vascular resistance (VR):

$$\Delta P = CO \times VR$$

Equation 1

The efficiency and performance of the heart (or LVAD) can be determined by the patient's cardiac pressure-volume (PV) loop. A cardiac PV loop plots the pressure and volume of the blood in the left ventricle throughout a single cardiac cycle (Figure 2).

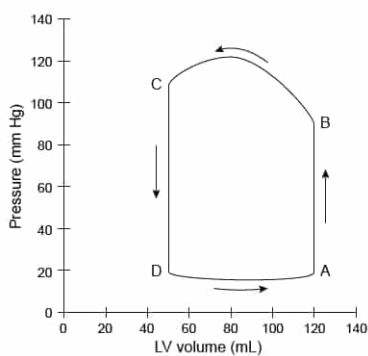


Figure 2 Cardiac PV Loop of a pulsatile-flow LVAD

Blood pressure is often represented by only two numbers: arterial systolic and diastolic pressures. A more detailed representation of blood pressures is shown in a blood pressure profile, which graphs the blood pressure throughout the length of the different vessel groups (Figure 3). Multiple pressure fluctuations are shown within a vessel because the pressure is traced across multiple cardiac cycles.

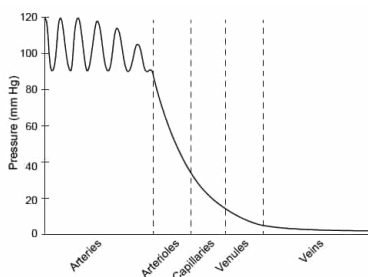


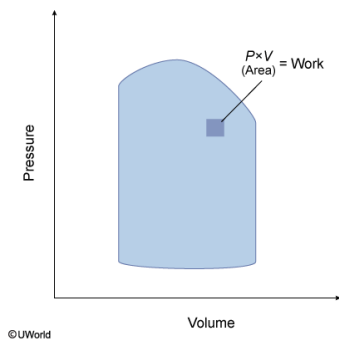
Figure 3 Blood pressure profile of the different vessels

The area enclosed by the cardiac PV loop is:

- ☐ A. the cardiac output. [19%]
- ☒ B. the work done. [70%]
- ☐ C. the power of the LVAD. [7%]
- ☐ D. the net pressure drop. [2%]

Correct
70% answered correctly

Explanation:



The physical quantity represented by an **area** on a graph is equivalent to the **product** of the **axes' quantities**. Therefore, the area on a cardiac PV loop represents the physical quantity equal to the product of pressure P and volume V .

$$[\text{area}] = PV$$

Pressure is a **derived quantity** ($P = F/A$) and can be **rewritten** as force divided by area.

$$PV = \frac{F}{A}V = \frac{FV}{A}$$

Volume has dimensions of $[\text{length}]^3$, and area has dimensions of $[\text{length}]^2$. Therefore, volume divided by area has dimensions of $[\text{length}]^3/[\text{length}]^2 = [\text{length}]$, or is distance d .

$$\frac{FV}{A} = F\frac{V}{A} = Fd$$

The amount of energy transferred by a force over a distance is the general definition of work: **$W = Fd$** . Because the product of pressure and volume is equal to the product of force and distance, the **area enclosed** by the cardiac PV loop is the amount of **work done**.

(Choice A) Cardiac output is the volume of blood pumped by the heart every minute and is the product of stroke volume and heart rate. Although stroke volume is represented by the width of the PV loop, heart rate is not represented by the height of the loop.

(Choice C) Power is the amount of work done over time ($P = W/t$). Although the area of the PV loop represents the amount of work done, time is not represented on a cardiac PV loop.

(Choice D) The net pressure drop is represented by the height of the PV loop.

Educational objective:

An area on a graph represents the physical quantity equal to the product of the axes' quantities. On a cardiac PV loop, an area represents pressure $\times$ volume and is equal to the amount of work involved.

Time spent: 0 sec
Subject:
 Physics

QID: 400821
System:
 Fluids

In the event of end-stage heart failure, a left ventricular assist device (LVAD) can be used as a heart transplant bridge to keep a patient alive. The first iterations of LVADs were pulsatile and mimicked the physiological pumping action of the heart (Figure 1).

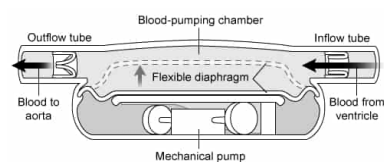


Figure 1 A pulsatile-flow LVAD schematic

A pulsatile-flow LVAD assists ventricular systole by mechanically pumping blood from a weakened left ventricle into the aorta through a pair of one-way valves. The pressure differential ΔP generated by the pump is related to cardiac output (CO) and vascular resistance (VR):

$$\Delta P = CO \times VR$$

Equation 1

The efficiency and performance of the heart (or LVAD) can be determined by the patient's cardiac pressure-volume (PV) loop. A cardiac PV loop plots the pressure and volume of the blood in the left ventricle throughout a single cardiac cycle (Figure 2).

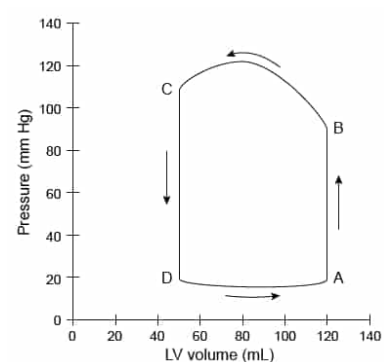


Figure 2 Cardiac PV Loop of a pulsatile-flow LVAD

Blood pressure is often represented by only two numbers: arterial systolic and diastolic pressures. A more detailed representation of blood pressures is shown in a blood pressure profile, which graphs the blood pressure throughout the length of the different vessel groups (Figure 3). Multiple pressure fluctuations are shown within a vessel because the pressure is traced across multiple cardiac cycles.

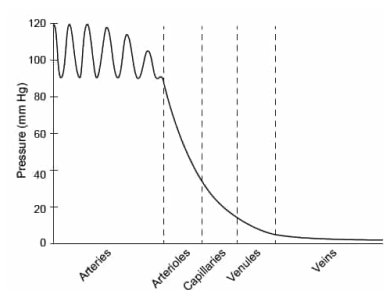
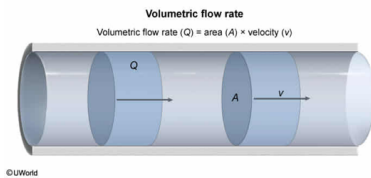


Figure 3 Blood pressure profile of the different vessels

If the radius of the outflow tube is 1 cm and the LVAD pump generates a pressure of 45 mmHg, what is the velocity of blood through the outflow tube if peripheral resistance is 1 mmHg·s·mL⁻¹?

- ✔ ☒ A. 45/π cm/s [44%]
- ☐ B. 45/π cm³/s [28%]
- ☐ C. 45 cm/s [15%]
- ☐ D. 45 cm³/s [11%]

Explanation:



The **volumetric flow rate** of blood through the outflow tube is determined by the heart (or LVAD) and is **equal to cardiac output**. From Equation 1, the pressure generated by the heart (ΔP) is the product of cardiac output (CO) and vascular resistance (VR):

$$\Delta P = CO \times VR$$

The CO of the LVAD is calculated using the given $\Delta P = 45 \text{ mmHg}$ and $VR = 1 \text{ mmHg} \cdot \text{s} \cdot \text{mL}^{-1}$,

$$CO = \frac{\Delta P}{VR}$$
$$CO = \frac{45 \text{ mmHg}}{1 \text{ mmHg} \cdot \text{s} \cdot \text{mL}^{-1}} = 45 \frac{\text{mL}}{\text{s}}$$

The volumetric flow rate $Q = CO = 45 \text{ mL/s}$. This value can be rewritten in **base SI units** ($1 \text{ mL} = 1 \text{ cm}^3$) as $Q = 45 \text{ cm}^3/\text{s}$. **Volumetric flow rate** is the product of cross-sectional area A and flow velocity v :

$$Q = Av$$

Because the cross-section of a tube is a circle ($A = \pi r^2$) with a radius $r = 1 \text{ cm}$, the outflow tube's cross-sectional area $A = \pi(1 \text{ cm})^2 = \pi \text{ cm}^2$. Flow velocity v is calculated using $Q = 45 \text{ cm}^3/\text{s}$ and $A = \pi \text{ cm}^2$:

$$v = \frac{Q}{A}$$
$$v = \frac{45 \text{ cm}^3/\text{s}}{\pi \text{ cm}^2} = \frac{45}{\pi} \text{ cm/s}$$

The velocity of blood through the outflow tube is **$45/\pi \text{ cm/s}$** .

(Choice B) Flow velocity is given in units of $[\text{length}]/[\text{time}]$ whereas volumetric flow rate is given in units of $[\text{length}]^3/[\text{time}]$.

(Choice D) $45 \text{ cm}^3/\text{s}$ is the volumetric flow rate (not flow velocity) in the outflow tube.

Educational objective:

The volumetric flow rate of blood in the outflow tube is equal to cardiac output (CO), which is the pressure (ΔP) generated by the heart divided by peripheral resistance (VR): $Q = CO = \Delta P/VR$. Flow velocity is determined from volumetric flow rate, which is the product of flow velocity and cross-sectional area: $Q = Av$.

Time spent: 0 sec
Subject:
Physics

QID: 400822
System:
Fluids

In the event of end-stage heart failure, a left ventricular assist device (LVAD) can be used as a heart transplant bridge to keep a patient alive. The first iterations of LVADs were pulsatile and mimicked the physiological pumping action of the heart (Figure 1).

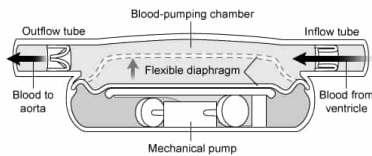


Figure 1 A pulsatile-flow LVAD schematic

A pulsatile-flow LVAD assists ventricular systole by mechanically pumping blood from a weakened left ventricle into the aorta through a pair of one-way valves. The pressure differential ΔP generated by the pump is related to cardiac output (CO) and vascular resistance (VR):

$$\Delta P = CO \times VR$$

Equation 1

The efficiency and performance of the heart (or LVAD) can be determined by the patient's cardiac pressure-volume (PV) loop. A cardiac PV loop plots the pressure and volume of the blood in the left ventricle throughout a single cardiac cycle (Figure 2).

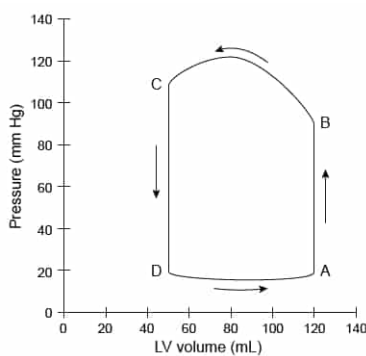


Figure 2 Cardiac PV Loop of a pulsatile-flow LVAD

Blood pressure is often represented by only two numbers: arterial systolic and diastolic pressures. A more detailed representation of blood pressures is shown in a blood pressure profile, which graphs the blood pressure throughout the length of the different vessel groups (Figure 3). Multiple pressure fluctuations are shown within a vessel because the pressure is traced across multiple cardiac cycles.

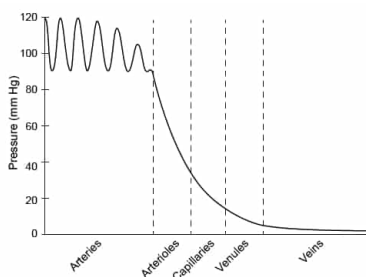


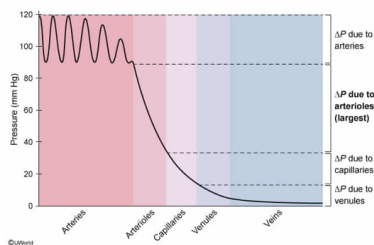
Figure 3 Blood pressure profile of the different vessels

Which group of blood vessels is the main source of peripheral resistance?

- ☐ A. The arteries, because they cause the greatest pressure fluctuations [21%]
- ☒ B. The arterioles, because they experience the largest pressure drop [43%]
- ☐ C. The capillaries, because they have the smallest vessel radii [28%]
- ☐ D. The venules, because they decrease blood pressure to its minimum [6%]

Correct
42% answered correctly

Explanation:



Equation 1 is an example of **Ohm law for flow**, which states that the pressure drop (ΔP) across a pipe (or vessel) is directly proportional to its resistance (R):

$$\Delta P = Q \times R$$

where Q is the volumetric flow rate. In other words, a relatively high pressure is required to drive flow through highly resistive vessels. Therefore, the group of vessels that has the **greatest pressure drop** across its entire length has the greatest resistance and is the **main source** of peripheral resistance. From the blood pressure profile of different vessels (Figure 3), the greatest pressure drop is seen across the **arterioles**.

In addition to having the greatest vascular resistance, the arterioles are also one of the main regulators of blood pressure. Unlike other vessels, the arterioles are very muscular and can significantly change their vessel radii. By constricting or relaxing, these vessels can

change the total peripheral resistance to maintain blood pressures as necessary.

(Choice A) These pressure fluctuations seen in the large arteries are not due to the resistance of the vessel. The pressure within the vessel fluctuates between its systolic and diastolic pressure depending on whether the heart is contracting or relaxing.

(Choice C) Although the capillaries have the smallest vessel radii and high *individual* resistances, they are the most numerous and are in parallel with each other. Therefore, the *equivalent resistance of the parallel network* of capillaries is relatively small.

(Choice D) Vascular resistance depends on the pressure drop across a vessel, not the pressure at its end.

Educational objective:

From Ohm law for flow ($\Delta P = Q \times R$), the pressure drop across a vessel is proportional to its resistance. Therefore, the group of vessels with the greatest pressure drop across its length, the arterioles, is the main source of peripheral resistance.

Time spent: 0 sec

Subject:
Physics

QID: 400823

System:
Fluids

In the event of end-stage heart failure, a left ventricular assist device (LVAD) can be used as a heart transplant bridge to keep a patient alive. The first iterations of LVADs were pulsatile and mimicked the physiological pumping action of the heart (Figure 1).

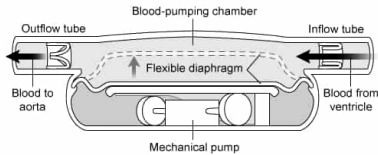


Figure 1 A pulsatile-flow LVAD schematic

A pulsatile-flow LVAD assists ventricular systole by mechanically pumping blood from a weakened left ventricle into the aorta through a pair of one-way valves. The pressure differential ΔP generated by the pump is related to cardiac output (CO) and vascular resistance (VR):

$$\Delta P = CO \times VR$$

Equation 1

The efficiency and performance of the heart (or LVAD) can be determined by the patient's cardiac pressure-volume (PV) loop. A cardiac PV loop plots the pressure and volume of the blood in the left ventricle throughout a single cardiac cycle (Figure 2).

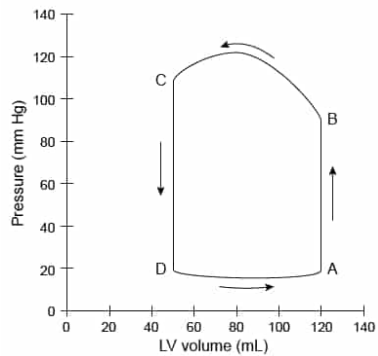


Figure 2 Cardiac PV Loop of a pulsatile-flow LVAD

Blood pressure is often represented by only two numbers: arterial systolic and diastolic pressures. A more detailed representation of blood pressures is shown in a blood pressure profile, which graphs the blood pressure throughout the length of the different vessel groups (Figure 3). Multiple pressure fluctuations are shown within a vessel because the pressure is traced across multiple cardiac cycles.

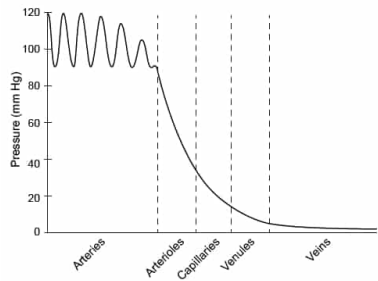
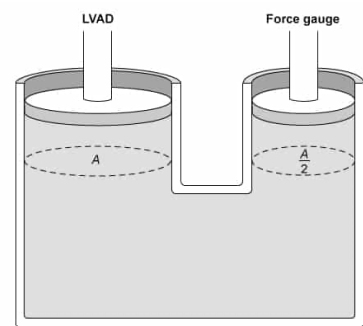


Figure 3 Blood pressure profile of the different vessels



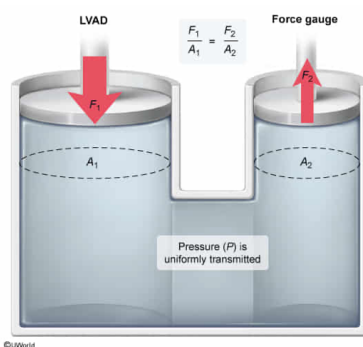
In the above figure, an LVAD pump is placed at one end of a "U"-shaped container, and a force gauge is placed at the other end. If the container is filled with an incompressible fluid, what is the ratio of the force measured by the force gauge to the force exerted by the LVAD?

- ☐ A. 1:4 [14%]
- ✓ ☒ B. 1:2 [58%]
- ☐ C. 1:1 [9%]
- ☐ D. 4:1 [16%]

Correct

57% answered correctly

Explanation:



For a static, incompressible (constant density), and fully enclosed fluid, an **external pressure** applied to the fluid will be uniformly **transmitted to all surfaces**. This principle is known as **Pascal law**. Therefore, the pressure applied by the LVAD pump in the figure is experienced by all surfaces of the fluid, including the force gauge.

Pressure P is a measure of a force F exerted over an area A :

$$P = \frac{F}{A}$$

Equating the pressure P_1 applied by the LVAD and the pressure P_2 experienced by the force gauge,

$$P_1 = P_2$$

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

For two equal pressures, a greater force will be exerted over a proportionally larger area (and vice versa). Rearranging the above equation as a ratio of the force F_2 exerted by LVAD to the force F_1 measured by the force gauge,

$$\frac{F_2}{F_1} = \frac{A_2}{A_1}$$

The **ratio of the forces** is **equal** to the **ratio of their respective areas**. According to the figure, the gauge area A_2 is half the LVAD area A_1 ($A_2 = A_1/2$). Therefore, the force experienced by the force gauge is 1/2 the LVAD force, corresponding to a **1:2** ratio between the force at the force gauge and the force at the LVAD

(Choices A and D) Because the pressures are equal and the force gauge area is smaller than the LVAD area, the force exerted on the force gauge must be proportionally smaller than the force exerted by the LVAD.

(Choice C) According to Pascal law, an externally applied *pressure* is uniformly exerted at all surfaces, not the force. The forces will be equal only if the areas are also equal.

Educational objective:

According to Pascal law, an external pressure exerted onto a static, incompressible, and fully enclosed fluid will be uniformly transmitted to all surfaces. Because pressure is measured as the force over an area, the ratio of the forces will be equal to the ratio of their respective areas.